Electric force: A force with magnitude and direction that acts between two charged particles.

Coulomb's Law: F_E is inversely proportional to the r^2
- an proportional to the product of q

$K = 8.99 \times 10^9 \ N \cdot m^2/C^2$

↳ Like charges: Repulsive
↳ unlike charges: Attractive
↳ The magnitude of F_E is always positive
↳ The (−) sign shows direction.
↳ Never include sign when solving Coulomb law problems
↳ The magnitude of the electric force is the magnitude of the force exerted on each of the particles
→ (Newton's third law: equal and opposite)

- Coulomb's Law and Universal Gravitation
↳ Both describe the force between 2 objects.
↳ Gravitation needs mass.
↳ Electric force needs electric charge
↳ Both decrease as distance between the 2 increases
↳ Gravitational forces are always attractive.
The direction of electric forces depend on ulikely or likely charges
The magnitude of electric force >>> magnitude of gravitational force.

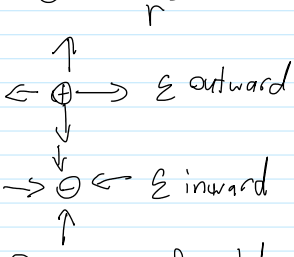
Superposition Principle: The total vector acting at a given point equals the sum of the individual vectors from all sources acting on the given point
↳ use component method and pytha

$$|F_E| = \frac{k|q_1||q_2|}{r^2}$$

An electric field creates electric force

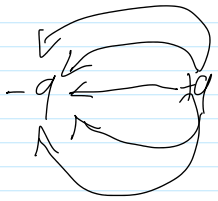
Electric field ($\vec{\mathcal{E}}$): The region in which a force is exerted on an electric charge. (N/C)
↳ It has a magnitude and direction.
↳ A charge creates an electric field
↳ A test charge located in the field will be affected by the existing electric field and experience an electric force
↳ $F_E = q\mathcal{E}$
↳ F_E is parallel to $\mathcal{E}$.
↳ F_E being (−) or (+) depends on the charge.
↳ for a positive test charge, $\mathcal{E}$ points in the same direction as the F_E acting on the test charge.
↳ for a (−) test charge, $\mathcal{E}$ points in the opp direction

$$\mathcal{E} = \frac{K|q_2|}{r^2}$$



Electric field lines: the continous lines of force around the charge that show direction

Electric dipoles: a pair of equal and opposite electric charges seperated by a small distance



The number of field lines for each charge is proportional to the magnitude of the charge

$E_E = -q\varepsilon\Delta d$

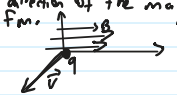
$\Delta V = \frac{\Delta E_E}{q}$

$\varepsilon = \frac{\Delta V}{\Delta d}$

ε = High potential energy → low potential energy

the source of electrical energy, a positive test charge will naturally move from the positive terminal where a high potential exists to the negative terminal where a low potential exists. Since the electric field points from positive to negative, the positive test charge will also move in the same direction as the field. Conversely, electrons will naturally travel from a region of low potential to a region of high potential, in a direction opposite to the direction of the electric field.

Right hand rule for a moving charge in a magnetic field
If you point your right thumb in the direction of the velocity of the charge ($\vec{v}$) and your straight fingers in the direction of the magnetic field (B). Your palm will point in the direction of the force



- The right-hand rule for a straight conductor states that if your thumb points in the direction of conventional current, your curled fingers indicate the direction of the magnetic field lines around the straight conductor.
- The right-hand rule for a solenoid states that if the fingers of your right hand curl in the direction of the conventional current, your thumb points in the direction of the magnetic field lines in the centre of the coil.